

**COURSE TITLE : TEXTILE WET PROCESSING I**  
**COURSE CODE : 3063**  
**COURSE CATEGORY : B**  
**PERIODS/WEEK : 4**  
**PERIODS/SEMESTER : 60**  
**CREDITS : 4**

### **TIME SCHEDULE**

| Module | Topics                                                  | Periods |
|--------|---------------------------------------------------------|---------|
| 1      | Chemistry of Wetting & Preparation of Cotton for Dyeing | 15      |
| 2      | Bleaching & Mercerising                                 | 15      |
| 3      | Introduction to Dyeing & Dyeing with water soluble Dyes | 15      |
| 4      | Dyeing with Water insoluble Dyes                        | 15      |
| TOTAL  |                                                         | 60      |

### **COURSE OUTCOME**

| Module | GO | Student will be able to                                                      |
|--------|----|------------------------------------------------------------------------------|
| 1      | 1  | Know the various chemicals used in textile wet processing.                   |
|        | 2  | Understand the various pre-treatments for wet processing of textiles         |
| 2      | 1  | Understand the bleaching process given to textile materials                  |
|        | 2  | Know the Mercerising process given to textile materials                      |
| 3      | 1  | Understand the process of dyeing cotton and viscose using water soluble dyes |
|        | 2  | Know the dyeing of wool and silk using water soluble dyes                    |
| 4      | 1  | Understand the process of dyeing of cotton using water insoluble dyes        |
|        | 2  | Know the process of dyeing of viscose using water insoluble dyes             |

### **SPECIFIC OUTCOME**

#### **MODULE I CHEMISTRY OF WETTING & PREPARATION OF COTTON FOR DYEING**

##### **1.1.0 Know the various chemicals used in textile wet processing**

- 1.1.1 List surface active agents & their classification.
- 1.1.2 Identify Textile processing chemicals
- 1.1.3 List some important commercial textile chemicals.
- 1.1.4 Explain the manufacture of soaps & detergents.

**1.2.0 Understand the various pre-treatments for wet processing of textiles**

- 1.2.1 Explain the sequence of operations in wet processing cotton Textiles.
- 1.2.2 State the objectives of singeing
- 1.2.3 Describe the methods of singeing.
- 1.2.4 State the objectives of de-sizing.
- 1.2.5 Explain the various methods of de-sizing.
- 1.2.6 List importance of scouring.
- 1.2.7 Describe scouring.
- 1.2.8 List types of keirs, explain the features & working of a modern Kier.

**MODULE II BLEACHING & MERCERISING**

**2.1.0 Understand the bleaching process given to textile materials**

- 2.1.1 List the objectives of bleaching
- 2.1.2 Identify various bleaching agents and compare.
- 2.1.3 State the principle involved in bleaching.
- 2.1.4 Explain the chemical behavior and application of calcium hypochlorite
- 2.1.5 Explain the preparation, properties and chemical behavior of hydrogen peroxide.
- 2.1.6 Describe the continuous method of bleaching of cotton using 'J' Box.

**2.2.0 Know the Mercerising process given to textile materials**

- 2.2.1 Define Mercerization & state its importance
- 2.2.2 Illustrate different parameters which affect mercerization.
- 2.2.3 Describe the different methods of mercerization.
- 2.2.4 Explain the working of a continuous cloth mercerizing machine.

**MODULE III INTRODUCTION TO DYEING & DYEING WITH WATER SOLUBLE DYES**

**3.1.0 Understand understand the process of dyeing cotton and viscose using water soluble colours**

- 3.1.1 Define dyes
- 3.1.2 State the classification of dyes
- 3.1.3 Differentiate dyeing and printing, compare dyes and pigments.
- 3.1.4 State the basic terms used in dyeing.
- 3.1.5 State the mechanism of dyeing.
- 3.1.6 Differentiate natural and synthetic dyes
- 3.1.7 List some important natural dyes.
- 3.1.8 Explain the method of dyeing Cotton and Viscose with Direct & Reactive dyes
- 3.1.9 Explain the method of application of Basic dyes on Cotton

**3.2.0 Know the dyeing of wool and silk using water soluble dyes**

- 3.1.1 Describe the method of application of Basic dyes on silk
- 3.1.2 Explain the application of Acid dyes on silk
- 3.1.3 Describe the method of application of Basic dyes on wool
- 3.1.2 Explain the application of Acid dyes on wool

## **MODULE IV DYEING WITH WATER INSOLUBLE DYES**

### **4.1.0 Understand the process of dyeing cotton using water insoluble dyes**

- 4.1.1 Describe the Dyeing of cotton with sulphur dyes
- 4.1.2 Explain the Dyeing of cotton with vat dyes
- 4.1.3 Describe the Dyeing of cotton with azoic dyes
- 4.1.4 Explain the Dyeing of cotton with oxidised dyes
- 4.1.5 Describe the Dyeing of cotton with mineral dyes

### **4.2.0 Know the process of dyeing viscose using water insoluble dyes**

- 4.2.1 Explain the Dyeing of viscose with vat dyes
- 4.2.2 Describe the Dyeing of viscose with azoic dyes

## **CONTENT DETAILS**

### **MODULE I**

#### **CHEMISTRY OF WETTING & PREPARATION OF COTTON FOR DYEING**

Definition of surface tension and contact angle- Relationship of surface tension and contact angle with wetting of textile materials- Important Surface active agents- Textile processing chemicals at a glance  
Brief study of manufacture of Soap & Detergents- Sequence of operations in wet processing of Cotton  
Textiles- Singeing- Objectives- Methods– Hot plate, Roller, Gas singeing machines- De-sizing- Objectives- Principle- Methods - Rot steeping- Acid steeping, Enzymatic de-sizing, Scouring- Objectives and importance - Principle and mechanism- Definition and types of Kiers- Features of a modern Kiers and its working- Scouring of coloured goods.

### **MODULE II**

#### **BLEACHING & MERCERISING**

Objectives of bleaching- Types of bleaching agents- Bleaching with hypochlorites- Comparison of sodium and calcium hypochlorites in terms of their properties, chemical behaviour and application- Method of bleaching with hypochlorites- Preparation of hydrogen peroxide- Properties and chemical behaviour of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage- Bleaching with Hydrogen peroxide (Batch process)- Continuous scouring and bleaching by using 'J' Box- Definition and principle of mercerizing- Changes in the appearance of cotton by mercerizing- Brief explanation of effect of following parameters, Concentration of NaOH, pH, Temperature- Time- Working of continuous Cloth Mercerizing machine- Working of a Hank Mercerising machine

## **MODULE III**

### **INTRODUCTION TO DYEING , DYES AND DYEING OF COTTON , VISCOSE, SILK & WOOL USING SOLUBLE D**

Brief history and development of colouring matters- Dye molecule- Differentiate dyes and pigments- Classification of dyes based on the method of application- General terms used in dyeing, Fastness, Material Liquor ratio & Dye bath- General mechanism of dyeing , Physical, Chemical & Solvent Theories of dyeing- Fastness properties of dyed materials- Important natural dyes- Differentiate natural dyes and synthetic dyes- Explain the method of application of Direct Dyes and Reactive Dyes on Cotton & Viscose - Basic colours on Cotton , wool & Silk - Acid colours on Wool& Silk.

## **MODULE IV**

### **DYEING OF COTTON AND VISCOSE USING WATER INSOLUBLE DYES AND NATURAL INDIGO**

Dyeing with Sulphur, Vat, Solubilised vat, Natural Indigo, Oxidized colours, Mineral and Azoic colours on cotton, dyeing of viscose with vat and azoic colours

## **REFERENCE BOOKS**

1. V.A. Shenai Technology of Textile Processing (Volume 1 to 5) Mahajan publishers  
Ahmadabad
2. E.R.Trotman Bleaching, Dyeing and Chemical Technology of Textile fibres  
Universal Publishing corp., Mumbai -2
3. R.S.Prayag Scouring, Bleaching, Mercerizing and Dyeing of Cellulose Fibres:
4. Srivastava Recent processes in Textile bleaching and dyeing
5. R.R. Chakravarthi A Glimpse on the chemical Technology of Textile fibres and types of  
Kiers-  
Features of a modern Kiers and its working- Scouring of coloured  
goods.
6. V.A. Shenai Bleaching, Dyeing and Chemical processing Technology of Textile fibres:  
E.R.Throtm